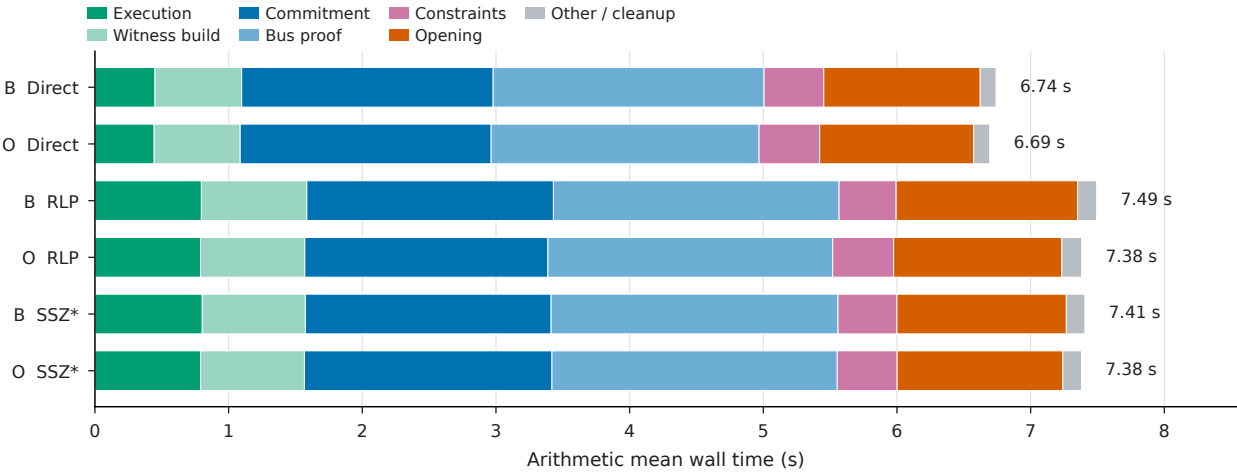
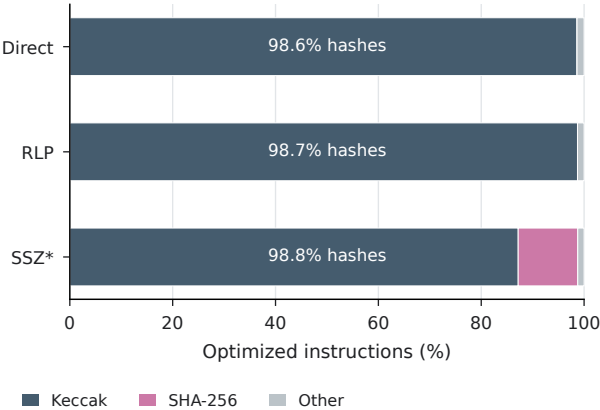


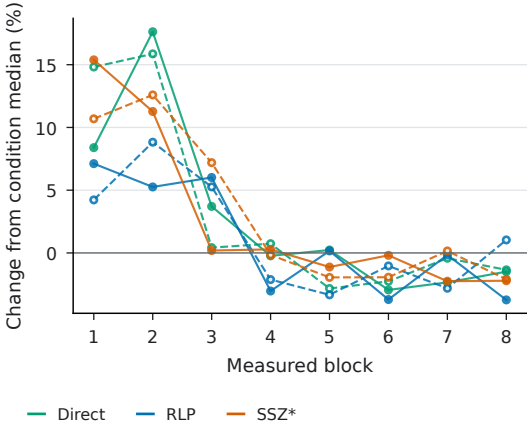
a Where complete proving time is spent



b Hashing dominates guest instructions



c Observe variation across the batch



Phase clocks are exclusive. Hash instruction counts are not an additive breakdown of proof-construction time.

B, baseline (dashed in c); O, optimized (solid). 8 fresh processes per condition; AC power; 11 Rayon workers.